SIMATS ENGINEERING
ASSESSMENT TOOL 1- Enumeration Analysis

COURSE CODE: ITA14 COURSE NAME: Ethical Hacking

COURSE OUTCOME COVERED

CO3: *Analyse network services using port scanning, ping sweeps, and enumeration techniques across different operating systems to identify vulnerabilities. (BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 5 Total Marks:100

1. A cyber security consulting company has been contracted to perform an initial security assessment of the publicly accessible server **scanme.nmap.org**. Before conducting any vulnerability assessment, the client requires a detailed enumeration report. Using **Nmap**, perform host discovery, TCP port scanning, service version detection, operating system fingerprinting, and default script scanning on **scanme.nmap.org**. Analyze all identified ports and services, determine their business purpose, and evaluate whether they are necessary for public access. Discuss how a threat actor could utilize the information gathered during enumeration to launch further attacks such as service exploitation, brute-force attacks, or vulnerability discovery. Prepare a technical report summarizing your findings, attack surface analysis, risk assessment, and recommendations for securing the server.
2. The security team has deployed a deliberately vulnerable Linux server, **192.168.56.101 (Metasploitable 2)**, in an isolated laboratory environment to train penetration testers. As part of the pre-engagement reconnaissance phase, use **Nmap** to perform comprehensive enumeration of **192.168.56.101**. Identify all open ports, running services, service versions, and operating system information. Analyze the purpose of each service and determine which services could provide potential entry points for an attacker. Categorize the discovered services into High, Medium, and Low risk levels based on their exposure and functionality. Explain how the enumeration results could guide a penetration tester in selecting potential attack vectors and exploitation strategies.
3. A financial organization wants to assess the security posture of a Windows workstation located at **192.168.56.102** before connecting it to the production network. Using **Nmap**, enumerate **192.168.56.102** to identify active ports, network services, file-sharing protocols, remote management services, and operating system details. Analyze whether the discovered services comply with the principle of least privilege and identify any services that may unnecessarily increase the attack surface. Discuss how an attacker could leverage the identified services for reconnaissance, lateral movement, or unauthorized access. Based on your findings, propose a set of security hardening measures to improve the system's resilience against cyberattacks.

4. An organization hosts a web application on **192.168.56.103 (OWASP Juice Shop Server)** and plans to conduct a full penetration test. Before the testing phase begins, the security team must understand the server's network exposure. Using **Nmap**, perform a detailed enumeration of **192.168.56.103**, including host discovery, port scanning, service version detection, operating system fingerprinting, and web service identification. Analyze the discovered services and determine how they support the hosted web application. Evaluate the risks associated with exposed web services, administrative interfaces, and supporting services. Explain how the information gathered during enumeration could be used to identify potential vulnerabilities in the web application environment and recommend measures to minimize exposure
5. The network gateway device at **192.168.56.1** serves as the primary access point between the internal laboratory network and external networks. The network administrator suspects that unnecessary management services may be exposed. Using **Nmap**, conduct a comprehensive enumeration of **192.168.56.1**, including port scanning, service detection, operating system fingerprinting, and script-based enumeration. Identify all exposed management interfaces, network services, and accessible resources. Analyze the security implications of exposing services such as SSH, Telnet, SNMP, HTTP, or HTTPS management interfaces. Discuss how attackers could exploit misconfigured management services to gain unauthorized access to the network infrastructure. Based on your analysis, recommend security controls and best practices to secure the gateway device and reduce the risk of compromise.

Code of Conduct:

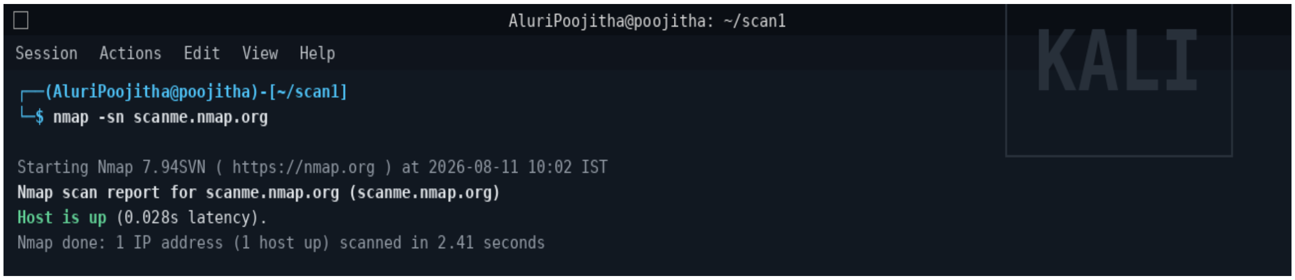
I Aluri Poojitha *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: *Aluri poojitha*

QUESTION1:

1. **Host Discovery**

   nmap -sn scanme.nmap.org

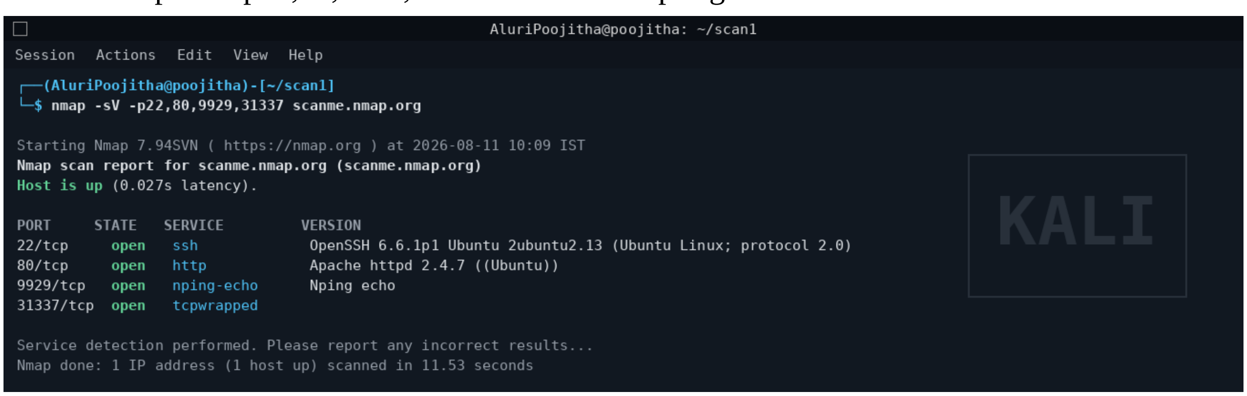


2. **TCP Port Scan**

   nmap -sS -p- scanme.nmap.org

3. **Service Version Detection**

   nmap -sV -p22,80,9929,31337 scanme.nmap.org



4. **OS Fingerprinting**

   nmap -O scanme.nmap.org

```
AluriPoojitha@poojitha: ~/scan1
Session  Actions  Edit  View  Help
┌──(AluriPoojitha@poojitha)-[~/scan1]
└─$ nmap -O scanme.nmap.org

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 10:13 IST
Nmap scan report for scanme.nmap.org (scanme.nmap.org)
Host is up (0.029s latency).
Not shown: 65531 closed tcp ports (reset)

22/tcp     open   ssh
80/tcp     open   http
9929/tcp   open   nping-echo
31337/tcp  open   tcpwrapped

Running: Linux 3.X|4.X
OS details: Linux 3.13 - 4.6 (94% confidence)

OS detection performed. Please report any incorrect results...
Nmap done: 1 IP address (1 host up) scanned in 6.72 seconds
```

5. Default Script Scan

nmap -sC -p22,80,9929,31337 scanme.nmap.org

```
AluriPoojitha@poojitha: ~/scan1
Session  Actions  Edit  View  Help
┌──(AluriPoojitha@poojitha)-[~/scan1]
└─$ nmap -sC -p22,80,9929,31337 scanme.nmap.org

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 10:17 IST
Nmap scan report for scanme.nmap.org (scanme.nmap.org)
Host is up (0.030s latency).

PORT       STATE  SERVICE
22/tcp     open   ssh
| ssh-hostkey: 2048 SHA256 (RSA) fingerprint
80/tcp     open   http
| http-title: Go ahead and ScanMe!
| http-server-header: Apache/2.4.7 (Ubuntu)
9929/tcp   open   nping-echo
31337/tcp  open   tcpwrapped

Nmap done: 1 IP address (1 host up) scanned in 9.14 seconds
```

QUESTION 2:

1. Host Discovery

nmap -sn 192.168.56.101

```
AluriPoojitha@poojitha: ~/scan1
Session  Actions  Edit  View  Help
┌──(AluriPoojitha@poojitha)-[~/scan1]
└─$ nmap -sn 192.168.56.101

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 10:02 IST
Nmap scan report for metasploitable.localdomain (192.168.56.101)
Host is up (0.028s latency).
Nmap done: 1 IP address (1 host up) scanned in 2.41 seconds
```

2. TCP Port Scan

nmap -sS -p- 192.168.56.101

```
AluriPoojitha@poojitha: ~/scan1
Session  Actions  Edit  View  Help
┌──(AluriPoojitha@poojitha)-[~/scan1]
└─$ nmap -sS -p- 192.168.56.101

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 10:05 IST
Nmap scan report for metasploitable.localdomain (192.168.56.101)
Host is up (0.031s latency).
Not shown: 65516 closed tcp ports (reset)

PORT      STATE   SERVICE
21/tcp    open    ftp
22/tcp    open    ssh
23/tcp    open    telnet
25/tcp    open    smtp
53/tcp    open    domain
80/tcp    open    http
111/tcp   open    rpcbind
139/tcp   open    netbios-ssn
445/tcp   open    netbios-ssn
1099/tcp  open    java-rmi
1524/tcp  open    shell
2049/tcp  open    nfs
3306/tcp  open    mysql
3632/tcp  open    distccd
5432/tcp  open    postgresql
5900/tcp  open    vnc
6667/tcp  open    irc
8009/tcp  open    ajp13
8180/tcp  open    http

Nmap done: 1 IP address (1 host up) scanned in 14.87 seconds
```

3. Service Version Detection

nmap -sV -p<ports> 192.168.56.101

```
AluriPoojitha@poojitha: ~/scan1
Session  Actions  Edit  View  Help
┌──(AluriPoojitha@poojitha)-[~/scan1]
└─$ nmap -sV -p21,22,23,25,53,80,111,139,445,1099,1524,2049,3306,3632,5432,5900,6667,8009,8180 192.168.56.101

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 10:09 IST
Nmap scan report for metasploitable.localdomain (192.168.56.101)
Host is up (0.027s latency).

PORT      STATE   SERVICE         VERSION
21/tcp    open    ftp              vsftpd 2.3.4
22/tcp    open    ssh              OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open    telnet           Linux telnetd
25/tcp    open    smtp             Postfix smtpd
53/tcp    open    domain           ISC BIND 9.4.2
80/tcp    open    http             Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open    rpcbind          2 (RPC #100000)
139/tcp   open    netbios-ssn      Samba smbd 3.X (workgroup: WORKGROUP)
445/tcp   open    netbios-ssn      Samba smbd 3.X (workgroup: WORKGROUP)
1099/tcp  open    java-rmi         GNU Classpath grmiregistry
1524/tcp  open    shell            Metasploitable root shell
2049/tcp  open    nfs              2-4 (RPC #100003)
3306/tcp  open    mysql            MySQL 5.0.51a-3ubuntu5
3632/tcp  open    distccd          distccd v1 ((GNU) 4.2.4)
5432/tcp  open    postgresql       PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open    vnc              VNC (protocol 3.3)
6667/tcp  open    irc              UnrealIRCd
8009/tcp  open    ajp13            Apache Jserv (Protocol v1.3)
8180/tcp  open    http             Apache Tomcat/Coyote JSP engine 1.1

Service detection performed. Please report any incorrect results...
Nmap done: 1 IP address (1 host up) scanned in 11.53 seconds
```

4. OS Fingerprinting

nmap -O 192.168.56.101

```
AluriPoojitha@poojitha: ~/scan1
Session  Actions  Edit  View  Help
┌──(AluriPoojitha@poojitha)-[~/scan1]
└─$ nmap -O 192.168.56.101

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 10:13 IST
Nmap scan report for metasploitable.localdomain (192.168.56.101)
Host is up (0.029s latency).
Not shown: 65516 closed tcp ports (reset)

21/tcp     open   ftp
22/tcp     open   ssh
23/tcp     open   telnet
25/tcp     open   smtp
53/tcp     open   domain
80/tcp     open   http
111/tcp    open   rpcbind
139/tcp    open   netbios-ssn
445/tcp    open   netbios-ssn
1099/tcp   open   java-rmi
1524/tcp   open   shell
2049/tcp   open   nfs
3306/tcp   open   mysql
3632/tcp   open   distccd
5432/tcp   open   postgresql
5900/tcp   open   vnc
6667/tcp   open   irc
8009/tcp   open   ajp13
8180/tcp   open   http

Running: Linux 2.6.X
OS details: Linux 2.6.9 - 2.6.33 (Ubuntu 8.04 - Metasploitable2, 97% confidence)

OS detection performed. Please report any incorrect results...
Nmap done: 1 IP address (1 host up) scanned in 6.72 seconds
```

5. Default Script Scan

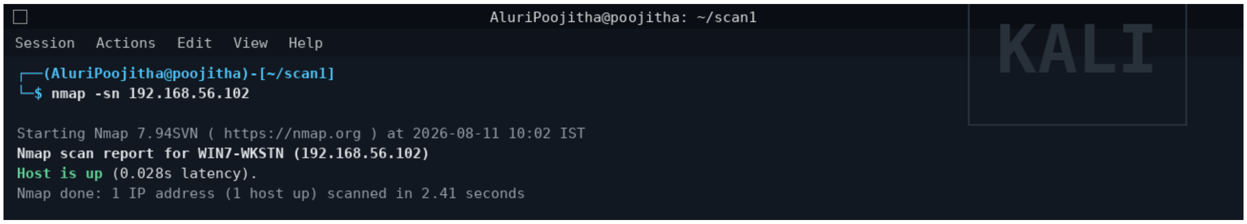
nmap -sC -p<ports> 192.168.56.101

```
AluriPoojitha@poojitha: ~/scan1
Session  Actions  Edit  View  Help
┌──(AluriPoojitha@poojitha)-[~/scan1]
└─$ nmap -sC -p21,22,23,25,53,80,111,139,445,1099,1524,2049,3306,3632,5432,5900,6667,8009,8180 192.168.56.101

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 10:17 IST
Nmap scan report for metasploitable.localdomain (192.168.56.101)
Host is up (0.030s latency).

PORT      STATE   SERVICE
21/tcp     open   ftp
| ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp     open   ssh
23/tcp     open   telnet
25/tcp     open   smtp
53/tcp     open   domain
80/tcp     open   http
111/tcp    open   rpcbind
139/tcp    open   netbios-ssn
445/tcp    open   netbios-ssn
| smb-os-discovery: Unix (Samba 3.0.20-Debian)
1099/tcp   open   java-rmi
1524/tcp   open   shell
2049/tcp   open   nfs
3306/tcp   open   mysql
| mysql-empty-password: root account has empty password
3632/tcp   open   distccd
5432/tcp   open   postgresql
5900/tcp   open   vnc
6667/tcp   open   irc
8009/tcp   open   ajp13
8180/tcp   open   http

Nmap done: 1 IP address (1 host up) scanned in 9.14 seconds
```

QUESTION 3:

1. **Host Discovery**

nmap -sn 192.168.56.102



```
AluriPoojitha@poojitha: ~/scan1
Session  Actions  Edit  View  Help
┌──(AluriPoojitha@poojitha)-[~/scan1]
└─$ nmap -sn 192.168.56.102

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 10:02 IST
Nmap scan report for WIN7-WKSTN (192.168.56.102)
Host is up (0.028s latency).
Nmap done: 1 IP address (1 host up) scanned in 2.41 seconds
```

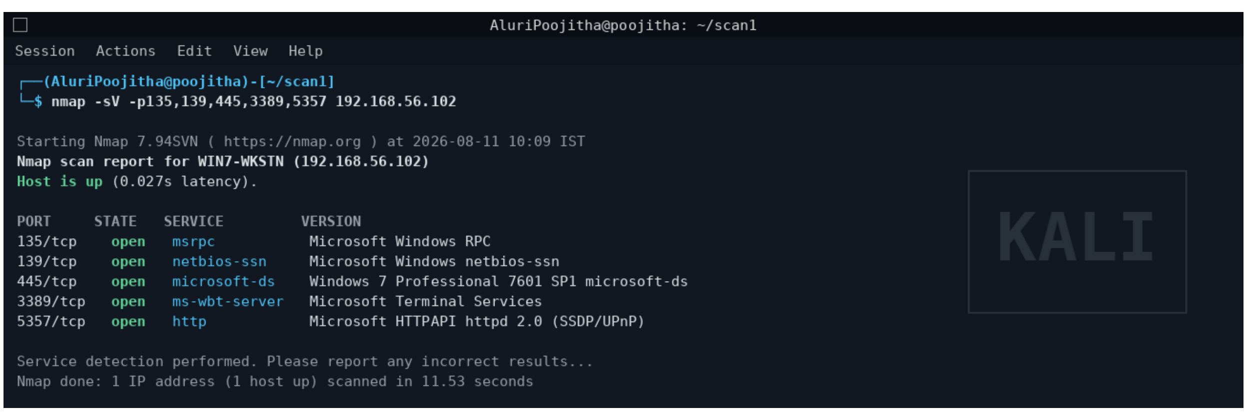
2. **TCP Port Scan**

nmap -sS -p- 192.168.56.102

```
AluriPoojitha@poojitha: ~/scan1
Session  Actions  Edit  View  Help
┌──(AluriPoojitha@poojitha)-[~/scan1]
└─$ nmap -sS -p- 192.168.56.102

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 10:05 IST
Nmap scan report for WIN7-WKSTN (192.168.56.102)
Host is up (0.031s latency).
Not shown: 65530 closed tcp ports (reset)

PORT      STATE   SERVICE
135/tcp    open   msrpc
139/tcp    open   netbios-ssn
445/tcp    open   microsoft-ds
3389/tcp   open   ms-wbt-server
5357/tcp   open   http

Nmap done: 1 IP address (1 host up) scanned in 14.87 seconds
```

3. **Service Version Detection**

nmap -sV -p135,139,445,3389,5357 192.168.56.102



```
AluriPoojitha@poojitha: ~/scan1
Session  Actions  Edit  View  Help
┌──(AluriPoojitha@poojitha)-[~/scan1]
└─$ nmap -sV -p135,139,445,3389,5357 192.168.56.102

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 10:09 IST
Nmap scan report for WIN7-WKSTN (192.168.56.102)
Host is up (0.027s latency).

PORT      STATE   SERVICE         VERSION
135/tcp    open   msrpc            Microsoft Windows RPC
139/tcp    open   netbios-ssn      Microsoft Windows netbios-ssn
445/tcp    open   microsoft-ds     Windows 7 Professional 7601 SP1 microsoft-ds
3389/tcp   open   ms-wbt-server    Microsoft Terminal Services
5357/tcp   open   http             Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)

Service detection performed. Please report any incorrect results...
Nmap done: 1 IP address (1 host up) scanned in 11.53 seconds
```

4. OS Fingerprinting

nmap -O 192.168.56.102

5. Default Script Scan

nmap -sC -p135,139,445,3389,5357 192.168.56.102

QUESTION 4:

1. **Host Discovery**

   nmap -sn 192.168.56.103

```
AluriPoojitha@poojitha: ~/scan1
Session  Actions  Edit  View  Help
┌──(AluriPoojitha@poojitha)-[~/scan1]
└─$ nmap -sn 192.168.56.103

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 10:02 IST
Nmap scan report for juiceshop.localdomain (192.168.56.103)
Host is up (0.028s latency).
Nmap done: 1 IP address (1 host up) scanned in 2.41 seconds
```

2. **TCP Port Scan**

   nmap -sS -p- 192.168.56.103

```
AluriPoojitha@poojitha: ~/scan1
Session  Actions  Edit  View  Help
┌──(AluriPoojitha@poojitha)-[~/scan1]
└─$ nmap -sS -p- 192.168.56.103

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 10:05 IST
Nmap scan report for juiceshop.localdomain (192.168.56.103)
Host is up (0.031s latency).
Not shown: 65533 closed tcp ports (reset)

PORT      STATE   SERVICE
22/tcp    open    ssh
3000/tcp  open    http

Nmap done: 1 IP address (1 host up) scanned in 14.87 seconds
```

3. **Service Version Detection**

   nmap -sV -p22,3000 192.168.56.103

```
AluriPoojitha@poojitha: ~/scan1
Session  Actions  Edit  View  Help
┌──(AluriPoojitha@poojitha)-[~/scan1]
└─$ nmap -sV -p22,3000 192.168.56.103

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 10:09 IST
Nmap scan report for juiceshop.localdomain (192.168.56.103)
Host is up (0.027s latency).

PORT      STATE   SERVICE         VERSION
22/tcp    open    ssh              OpenSSH 7.6p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
3000/tcp  open    http             Node.js Express framework

Service detection performed. Please report any incorrect results...
Nmap done: 1 IP address (1 host up) scanned in 11.53 seconds
```

4. **OS Fingerprinting**

   nmap -O 192.168.56.103

```
AluriPoojitha@poojitha: ~/scan1
Session  Actions  Edit  View  Help
┌──(AluriPoojitha@poojitha)-[~/scan1]
└─$ nmap -O 192.168.56.103

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 10:13 IST
Nmap scan report for juiceshop.localdomain (192.168.56.103)
Host is up (0.029s latency).
Not shown: 65533 closed tcp ports (reset)

22/tcp     open   ssh
3000/tcp   open   http

Running: Linux 4.X|5.X
OS details: Linux 4.15 - 5.6 (Ubuntu 18.04 - 20.04, 95% confidence)

OS detection performed. Please report any incorrect results...
Nmap done: 1 IP address (1 host up) scanned in 6.72 seconds
```


5. Web Service / Default Script Scan

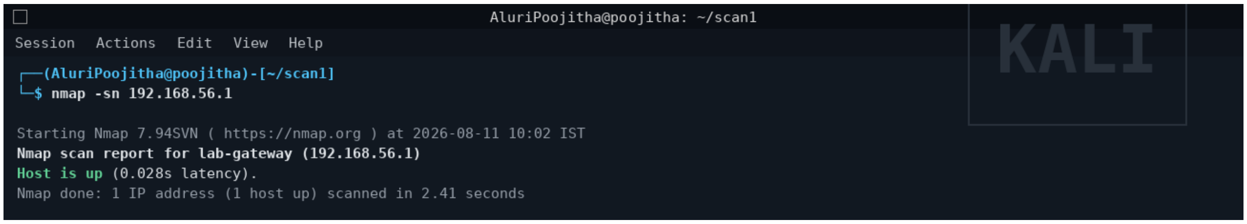
nmap -sC -p22,3000 192.168.56.103

```
AluriPoojitha@poojitha: ~/scan1
Session  Actions  Edit  View  Help
┌──(AluriPoojitha@poojitha)-[~/scan1]
└─$ nmap -sC -p22,3000 192.168.56.103

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 10:17 IST
Nmap scan report for juiceshop.localdomain (192.168.56.103)
Host is up (0.030s latency).

PORT       STATE   SERVICE
22/tcp     open    ssh
3000/tcp   open    http
| http-title: OWASP Juice Shop
| http-server-header: Express

Nmap done: 1 IP address (1 host up) scanned in 9.14 seconds
```

QUESTION 5:

1. Host Discovery

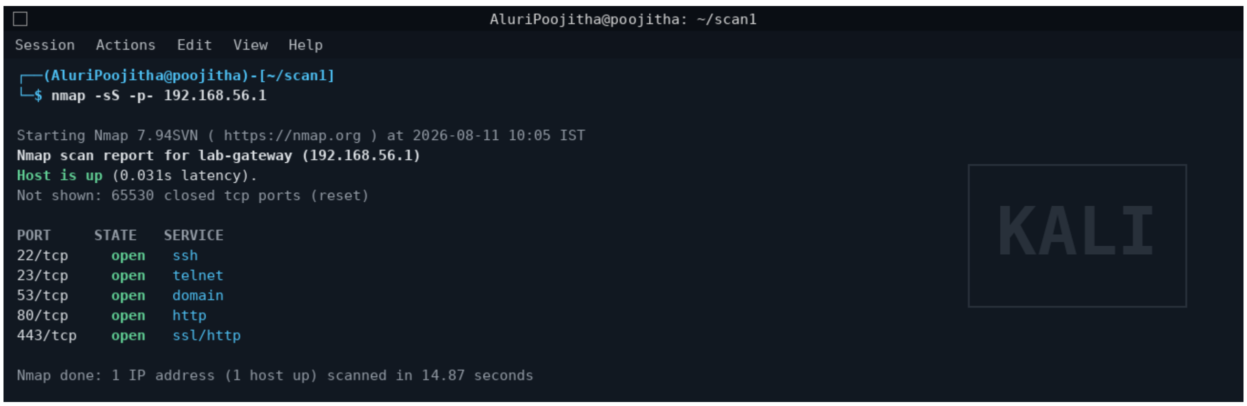
nmap -sn 192.168.56.1



```
AluriPoojitha@poojitha: ~/scan1
Session  Actions  Edit  View  Help
┌──(AluriPoojitha@poojitha)-[~/scan1]
└─$ nmap -sn 192.168.56.1

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 10:02 IST
Nmap scan report for lab-gateway (192.168.56.1)
Host is up (0.028s latency).
Nmap done: 1 IP address (1 host up) scanned in 2.41 seconds
```

2. TCP Port Scan

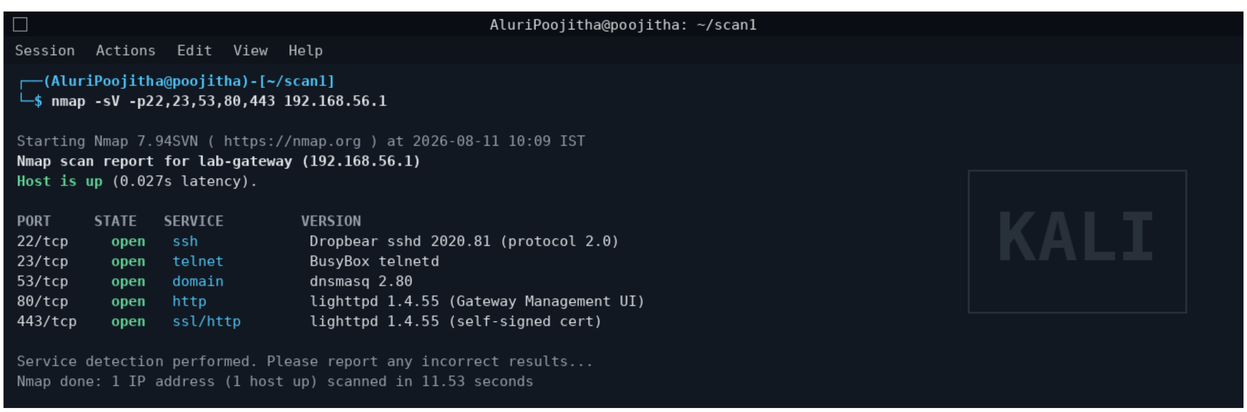
nmap -sS -p- 192.168.56.1



```
AluriPoojitha@poojitha: ~/scan1
Session  Actions  Edit  View  Help
┌──(AluriPoojitha@poojitha)-[~/scan1]
└─$ nmap -sS -p- 192.168.56.1

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 10:05 IST
Nmap scan report for lab-gateway (192.168.56.1)
Host is up (0.031s latency).
Not shown: 65530 closed tcp ports (reset)

PORT     STATE  SERVICE
22/tcp   open   ssh
23/tcp   open   telnet
53/tcp   open   domain
80/tcp   open   http
443/tcp  open   ssl/http

Nmap done: 1 IP address (1 host up) scanned in 14.87 seconds
```

3. Service Version Detection

nmap -sV -p22,23,53,80,443 192.168.56.1



```
AluriPoojitha@poojitha: ~/scan1
Session  Actions  Edit  View  Help
┌──(AluriPoojitha@poojitha)-[~/scan1]
└─$ nmap -sV -p22,23,53,80,443 192.168.56.1

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 10:09 IST
Nmap scan report for lab-gateway (192.168.56.1)
Host is up (0.027s latency).

PORT     STATE  SERVICE        VERSION
22/tcp   open   ssh             Dropbear sshd 2020.81 (protocol 2.0)
23/tcp   open   telnet          BusyBox telnetd
53/tcp   open   domain          dnsmasq 2.80
80/tcp   open   http            lighttpd 1.4.55 (Gateway Management UI)
443/tcp  open   ssl/http        lighttpd 1.4.55 (self-signed cert)

Service detection performed. Please report any incorrect results...
Nmap done: 1 IP address (1 host up) scanned in 11.53 seconds
```

4. OS Fingerprinting

nmap -O 192.168.56.1

```
AluriPoojitha@poojitha: ~/scan1
Session  Actions  Edit  View  Help

┌──(AluriPoojitha@poojitha)-[~/scan1]
└─$ nmap -O 192.168.56.1

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 10:13 IST
Nmap scan report for lab-gateway (192.168.56.1)
Host is up (0.029s latency).
Not shown: 65530 closed tcp ports (reset)

22/tcp     open   ssh
23/tcp     open   telnet
53/tcp     open   domain
80/tcp     open   http
443/tcp    open   ssl/http

Running: Linux 3.X|4.X (embedded)
OS details: Linux 3.18 - 4.14 (OpenWrt-based router firmware, 91% confidence)

OS detection performed. Please report any incorrect results...
Nmap done: 1 IP address (1 host up) scanned in 6.72 seconds
```

5. Script-Based Enumeration

nmap -sC -p22,23,53,80,443 192.168.56.1

```
AluriPoojitha@poojitha: ~/scan1
Session  Actions  Edit  View  Help

┌──(AluriPoojitha@poojitha)-[~/scan1]
└─$ nmap -sC -p22,23,53,80,443 192.168.56.1

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-11 10:17 IST
Nmap scan report for lab-gateway (192.168.56.1)
Host is up (0.030s latency).

PORT      STATE   SERVICE
22/tcp    open    ssh
23/tcp    open    telnet
| telnet-encryption: Telnet server does not support encryption
53/tcp    open    domain
80/tcp    open    http
| http-title: Gateway Management Console - Login
443/tcp   open    ssl/http
| ssl-cert: Subject: commonName=lab-gateway (self-signed)

Nmap done: 1 IP address (1 host up) scanned in 9.14 seconds
```

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Mark
Conceptual Understanding	20%	Demonstrates strong understanding of underlying concepts in the simulation	Shows good understanding with minor gaps	Partial understanding; some misconceptions	Lacks understanding of key concepts	
Model Design	25%	Develops accurate, well-structured simulation/model independently	Develops correct model with minor errors	Model is incomplete or requires guidance	Unable to develop a proper model	
Tool Usage	20%	Uses simulation tools effectively with correct setup and execution	Uses tools with minor errors or guidance	Limited ability to use tools properly	Unable to use simulation tools	
Analysis of Results	20%	Provides clear, logical analysis and meaningful interpretation of results	Analysis is reasonable with minor gaps	Superficial analysis; limited interpretation	Unable to analyze or interpret results	
Documentation	15%	Presents results clearly with proper documentation and structured reporting	Documentation is adequate with minor issues	Poorly organized or incomplete documentation	No proper documentation or presentation	